Retrieval-Augmented Generation for Large Language Models: A Survey

Retrieval-augmented generation (RAG) connects large language models to external evidence by retrieving documents, passages, memories, database records, or multimodal items and conditioning generation on the selected context. This survey reviews RAG for LLMs as a full pipeline rather than a single retrieval call. We synthesize an arXiv-derived corpus of 300 papers collected during this run and organize the literature around corpus construction, retrievers, query rewriting, reranking, evidence fusion, generator conditioning, attribution, evaluation, adaptation, domain-specific systems, and safety. The survey argues that RAG is best understood as a controllable interface between parametric knowledge and non-parametric evidence. Its strengths are updateability, provenance, and domain grounding; its weaknesses are retrieval misses, distractor sensitivity, stale corpora, context overload, and difficult end-to-end evaluation. We close with prospects for adaptive, auditable, and privacy-aware RAG systems.

Keywords:

large language models; retrieval-augmented generation; dense retrieval; grounding; factuality; evaluation

1 Introduction

Retrieval-augmented generation (RAG) has become one of the central patterns for making large language models more grounded, updateable, and domain-aware. Instead of relying only on parametric knowledge learned during pretraining, a RAG system retrieves external evidence and conditions the model's response on that evidence. This design is attractive because the corpus can be updated, inspected, filtered, cited, and specialized for a task without retraining the entire model [32, 90, 92, 102, 131, 267, 293, 299]. At the same time, RAG introduces a new systems problem: the final answer depends on the quality of the corpus, retriever, ranker, context assembler, generator, and verifier.

RAG is often described as retrieve-then-generate, but modern systems are more varied. Some rewrite the user's query, perform iterative or multi-hop search, rerank passages, compress evidence, ask the generator to cite sources, or use feedback to decide whether more retrieval is needed [32, 81, 92, 102, 131, 154, 267, 291]. Others blend retrieval with long-context modeling, agent memory, tool use, structured databases, knowledge graphs, or multimodal stores. The design space is therefore much broader than a vector database attached to a chatbot.

This survey treats RAG as an evidence interface between language models and external knowledge. A useful RAG system must answer four questions: what evidence is available, what evidence is relevant, how evidence should be represented in the prompt or model state, and how the generated answer should be checked against the evidence. These questions connect information retrieval, natural-language generation, databases, human-computer interaction, evaluation, and safety.

2 Background

2.1 Parametric and Non-Parametric Knowledge

LLMs store a large amount of knowledge in parameters, but that knowledge is difficult to update, inspect, or attribute. RAG adds non-parametric knowledge in the form of documents, indexes, tables, memories, graphs, or other external stores. The result is a hybrid system: the model contributes linguistic generalization and reasoning, while retrieval contributes fresh evidence and domain specificity. The balance is delicate. If retrieval misses key evidence, generation may hallucinate; if retrieval returns distractors, generation may overfit to irrelevant context; if too much evidence is inserted, the model may fail to use the right passages.

2.2 The RAG Pipeline

A typical RAG pipeline includes ingestion, chunking, indexing, query construction, retrieval, reranking, prompt assembly, generation, citation, and evaluation. Each stage encodes assumptions. Chunking determines what semantic units can be retrieved; embeddings determine similarity; query rewriting determines recall; reranking determines precision; and prompt assembly determines what the model can attend to. End-to-end quality is bounded by the weakest stage, which is why many recent studies evaluate full pipelines rather than isolated retrievers [15, 32, 93, 102, 154, 244, 285, 299].

2.3 Grounding and Attribution

Grounding requires that generated claims be supported by retrieved evidence. Attribution goes further by exposing which source supports which claim. These goals are important because RAG is often deployed in settings where users expect traceability: science, medicine, law, finance, enterprise search, and technical support. Yet citations can be misleading when the answer cites a retrieved passage that is topically related but does not actually entail the claim. Grounded RAG therefore needs claim-level verification, evidence selection, and calibration [34, 92, 93, 102, 131, 267, 288, 299].

3 Methods

3.1 Search Protocol and Corpus Construction

The corpus for this survey was generated during this run from the arXiv structured API. Queries combined the topic with retrieval-augmented generation, RAG, dense retrieval, document retrieval, question answering, grounding, hallucination, query rewriting, reranking, multi-hop retrieval, evaluation, benchmarks, long context, memory, multimodal retrieval, and domain-specific RAG terms. Records were retained when title or abstract evidence connected LLMs or language models with retrieval, external evidence, generation, grounding, evaluation, training, or safety. Duplicates were removed by arXiv identifier, and the final bibliography cites 300 unique scholarly preprints.

3.2 Taxonomy

We organize RAG methods along six axes. The first is *corpus design*: document sources, freshness, chunking, metadata, and access control. The second is *retrieval*: sparse, dense, hybrid, graph-based, structured, or multimodal search. The third is *evidence selection*: query rewriting, decomposition, reranking, filtering, and compression. The fourth is *generation*: how retrieved evidence is fused, cited, and reconciled with model knowledge. The fifth is *learning*: whether retrievers, rerankers, generators, or verifiers are trained jointly or separately. The sixth is *governance*: safety, privacy, provenance, and auditability.

3.3 Retrievers, Indexes, and Corpus Construction

Retrieval quality starts before any query is issued. The corpus must be cleaned, segmented, enriched with metadata, and indexed in a way that matches expected questions. Dense retrievers improve semantic recall, sparse retrievers preserve lexical precision, and hybrid systems exploit both. Corpus construction also shapes failure modes: overly small chunks lose context, overly large chunks reduce precision, stale indexes return outdated information, and missing metadata makes filtering difficult. These issues motivate work on adaptive chunking, hierarchical retrieval, hybrid search, and domain-specific indexing [15, 32, 93, 102, 154, 244, 285, 299].

3.4 Query Rewriting, Multi-Hop Retrieval, and Reranking

User questions are rarely optimal retrieval queries. Query rewriting expands, decomposes, or clarifies the information need before retrieval. Multi-hop RAG retrieves evidence iteratively, where earlier passages guide later queries. Rerankers then estimate which candidate passages are most useful for generation. These components are crucial because the generator is sensitive to both missing evidence and distractors. The best RAG systems treat evidence selection as an active process rather than a static top-k lookup [32, 81, 92, 102, 131, 154, 267, 291].

3.5 Generation, Fusion, and Answer Control

After retrieval, the model must fuse evidence with its own prior knowledge. Generation strategies vary from simple prompting to structured answer plans, citation-aware decoding, chain-of-thought over evidence, and verifier-guided revision. A central tension is that the model may know facts not present in the retrieved passages; using such knowledge can improve completeness but weaken attribution. RAG systems therefore need policies for abstention, conflict handling, source preference, and answer calibration [32, 90, 92, 102, 131, 267, 293, 299].

3.6 Training and Optimization

RAG components can be optimized separately or jointly. Retriever training improves recall, reranker training improves precision, generator tuning improves evidence use, and preference or reinforcement signals can reward faithful answers. End-to-end optimization is difficult because retrieval decisions are discrete, corpora evolve, and answer quality depends on both evidence and generation. Recent work explores synthetic data, self-training, contrastive learning, feedback loops, and verifier-based objectives [32, 34, 76, 93, 102, 131, 154, 299].

3.7 Evaluation and Diagnostics

RAG evaluation should distinguish retrieval quality from generation quality. A system may retrieve the right passage and still hallucinate, or retrieve weak evidence and answer correctly from parametric memory. Useful metrics include recall, answer correctness, citation faithfulness, context precision, robustness to distractors, abstention quality, latency, and cost. Benchmarks increasingly include multi-hop questions, long-context settings, domain corpora, and adversarial retrieval conditions [32, 76, 90, 102, 131, 267, 293, 299].

3.8 Domain-Specific and Multimodal RAG

Domain RAG systems adapt the pipeline to specialized evidence. Medical, legal, scientific, financial, code, table, graph, and multimodal settings differ in terminology, provenance requirements, latency, and risk. For example, biomedical RAG may prioritize controlled vocabularies and evidence hierarchy, while code RAG needs repository structure and executable verification. Multimodal RAG adds image, video, audio, and layout-aware retrieval, expanding the evidence interface beyond plain text.

3.9 Representative Literature Map

The following map cites the full retained corpus by theme, making the survey coverage explicit.

Retrievers, indexes, and corpus construction The retrievers, indexes, and corpus construction cluster contains work that treats LLMs as controllers of typed resources, stateful environments, or evaluation tasks [32, 68, 74, 106, 190, 191, 203, 204, 210]. A second strand in this cluster studies how such systems represent intermediate state, recover from failed calls, and compare tool-mediated behavior across datasets [61, 147, 171, 180, 213, 264].

Augmentation architectures and generation control The augmentation architectures and generation control cluster contains work that treats LLMs as controllers of typed resources, stateful environments, or evaluation tasks [76, 90, 92, 102, 131, 154, 267, 293, 299]. A second strand in this cluster studies how such systems represent intermediate state, recover from failed calls, and compare tool-mediated behavior across datasets [11, 34, 93, 165, 197, 244, 252, 254, 288]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [81, 107, 152, 159, 209, 221, 247, 285, 291]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [4, 12, 15, 49, 114, 115, 199, 217, 263]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [2, 30, 31, 60, 158, 200, 212, 266, 277]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [9, 62, 127, 142, 153, 214, 259, 298, 300]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [1, 13, 94, 129, 161, 219, 222, 240, 283]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [14, 85, 118, 120, 135, 193, 265, 279, 287]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [26, 113, 136, 145, 223, 224, 230, 237, 272]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [28, 66, 108, 116, 133, 179, 211, 248, 261]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [27, 37, 56, 70, 72, 103, 218, 243, 289]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [20, 21, 58, 77, 155, 163, 189, 257, 297]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [3, 7, 59, 96, 101, 104, 141, 229, 286]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [6, 42, 69, 97, 140, 183, 228, 238, 250]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [54, 64, 117, 121, 168, 170, 182, 227, 258]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [65, 79, 91, 119, 125, 172, 231, 235, 274]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [48, 73, 84, 148, 162, 192, 256, 276, 280]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [110, 128, 130, 143, 201, 215, 225, 251, 273]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [35, 38, 41, 55, 150, 176, 239, 292, 295]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [8, 53, 86, 100, 126, 177, 198, 226, 241]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [18, 33, 44, 109, 124, 164, 169, 234, 246]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [22, 43, 47, 87, 95, 98, 146, 262, 296]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [24, 29, 112, 134, 137, 157, 195, 206, 290]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [19, 67, 82, 89, 166, 173, 220, 233, 275]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [17, 23, 39, 57, 149, 156, 186, 196, 282]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [40, 63, 99, 122, 151, 178, 232, 236, 260]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [25, 71, 105, 144, 185, 187, 205, 216, 270]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [5, 10, 36, 46, 50, 83, 139, 184, 242]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [78, 194, 202].

Query rewriting, reranking, and evidence selection The query rewriting, reranking, and evidence selection cluster contains work that treats LLMs as controllers of typed resources, stateful environments, or evaluation tasks [52, 160].

Training, adaptation, and optimization The training, adaptation, and optimization cluster contains work that treats LLMs as controllers of typed resources, stateful environments, or evaluation tasks [188, 208].

Benchmarks, evaluation, and diagnostics The benchmarks, evaluation, and diagnostics cluster contains work that treats LLMs as controllers of typed resources, stateful environments, or evaluation tasks [16, 80, 167, 174, 175, 253, 268, 281, 284]. A second strand in this cluster studies how such systems represent intermediate state, recover from failed calls, and compare tool-mediated behavior across datasets [45, 181, 207, 245, 249, 255, 269, 271, 278]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [51, 75, 88, 111, 123, 132, 294].

Domain-specific and multimodal rag The domain-specific and multimodal rag cluster contains work that treats LLMs as controllers of typed resources, stateful environments, or evaluation tasks [138].

4 Challenges and Prospects

4.1 Retrieval Misses and Distractor Sensitivity

RAG systems can fail silently when they retrieve plausible but insufficient evidence. Increasing k improves recall but also increases distractors and context cost. Future systems need adaptive retrieval policies that decide when to search again, when to ask for clarification, and when to abstain.

4.2 Faithfulness and Citation Quality

Citation is not the same as support. A generated claim may cite a passage that contains related words but does not entail the claim. Better RAG systems will need claim-level attribution, entailment checks, and interfaces that expose uncertainty rather than presenting citations as decorative references [34, 92, 93, 102, 131, 267, 288, 299].

4.3 Freshness, Privacy, and Security

External corpora create governance obligations. Documents may become stale, confidential, poisoned, or inappropriate for a user's permission scope. RAG systems therefore need corpus versioning, access control, privacy filters, and monitoring for prompt injection and retrieval poisoning [4, 30, 32, 62, 212, 244, 266, 281].

4.4 Long-Context RAG and Memory

Long-context models change RAG but do not remove it. They can ingest more evidence, but retrieval remains valuable for freshness, filtering, provenance, and cost control. The future likely belongs to hybrid systems that combine long windows, hierarchical retrieval, memory stores, and evidence compression.

5 Conclusion

RAG has evolved from a simple retrieval-plus-generation recipe into a broad architecture for grounding LLMs in external evidence. This survey reviewed corpus construction, retrievers, rerankers, evidence fusion, training, evaluation, domain-specific systems, and safety. The central lesson is that RAG quality is pipeline quality: reliable answers require good evidence selection, faithful generation, transparent attribution, and governed corpora. Future RAG systems should be adaptive, auditable, privacy-aware, and explicit about uncertainty.

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